

ED Occupancy Forecasting (ML LoS) — Limitations & Remediation (One-Pager)

Context

Two components of the project: (A) ML prediction of individual ED patient Length-of-Stay (LoS) aggregated to forecast hospital-wide occupancy; (B) rare-cohort LoS improvement via CTGAN synthetic data. Both are trained/evaluated on MIMIC-IV ED discharges.

Key Limitations

A. LoS → Occupancy aggregation

- **Artificial timeline & no seasonality:** De-identified dates break day-of-week/seasonal effects; occupancy is computed from the same LoS used to build it (tautology risk).
- **Future arrivals handled with simple averages:** Uses constant averages for $t+1 \dots t+3$ cohorts; ignores calendar effects and case-mix drift.
- **Granularity mismatch:** Forecasts hospital-wide beds, not ward/acuity/isolation/sex-segregated needs.
- **Tail under-representation:** Very long stays truncated; these dominate peak occupancy.
- **Calibration gap:** Simulates “symmetry”/lower error rather than training for it; no reliability/quantile calibration reported.
- **No fairness audit:** Inputs (e.g., pain score, acuity) can encode bias; subgroup performance unknown.

B. Rare-cohort synthetic augmentation

- **Helps the not-so-rare:** CTGAN skipped or weak when cohorts $\lesssim 30$ cases—often the ones we most need improved.
- **Synthetic overweighting risk:** Fixed +4k synthetic per cohort can distort distributions (mode collapse / long-stay inflation).
- **Utility proxy, not outcome:** Heuristic realism scores aren’t tied to downstream LoS gains.
- **Privacy blindspot:** No leakage tests (membership/attribute inference) for synthetic data.
- **Narrow baselines:** Limited comparison beyond CatBoost; no survival/quantile/hierarchical models assessed.

Cross-cutting risks

- **Flow dynamics absent:** No queueing/boarding/discharge feedback loops; occupancy affects LoS and vice-versa.
- **Offline only:** No prospective/live evaluation or impact on decisions (e.g., elective smoothing, diversion avoidance).
- **Metric misalignment:** MAE on beds isn’t decision-centric; asymmetry costs (over vs under) not modeled.

Remediation Plan (concrete actions)

R1. Calendar-aware arrivals & stays

- Build arrival model with *dow/season/holiday* features and ED case-mix; pair with survival/quantile LoS models.
- Replace constant averages with sampling from these distributions for future cohorts.

R2. Train for aggregation quality

- Optimize LoS with quantile / pinball or asymmetric loss; add **post-hoc isotonic** calibration; explicitly minimize *occupancy MAE@Horizon* via differentiable aggregation or bilevel training.

R3. Granular forecasts

- Produce **ward/acuity/isolation/sex**-specific occupancy and required nurse skill-mix; incorporate bed blocks and cohorting constraints.

R4. Tail-aware modeling

- Include long-stay tails via AFT/survival mixtures; evaluate P95/P99 occupancy error and “overflow days wrong.”

R5. Fairness, safety, and governance

- Report subgroup error & calibration (sex, age, race/ethnicity, language, arrival mode, insurance).
- Add **guardrail rules** (e.g., max influence of pain score) and SHAP-based drift monitoring; convene clinical + ethics review.

R6. Synthetic data discipline

- Cap synthetic:real ratio (e.g., $\leq 1:1$); *utility-driven selection* (keep only synthetic samples that improve validation).
- Run privacy audits (nearest-neighbor distance, membership/attribute inference).
- Prefer DP-CTGAN for external data sharing; document provenance.

R7. Flow & policy integration

- Couple predictions with a simple **queueing/boarding** model and discharge scheduling; run what-if scenarios (escalation protocols, elective smoothing).

R8. Prospective pilot

- Shadow-mode for 4–6 weeks; pre-register decision metrics; then A/B rollout on scheduling and escalation huddles.

Decision-centric Metrics to Track

- **Beds MAE@H** (H = 24/48/72h) per ward + P95 error; **overflow/shortfall days**; mean signed error (bias).
- **Actionable gains**: elective cases scheduled, cancellations avoided, diversion hours reduced, boarding time.

- **Fairness:** error & calibration parity across subgroups; impact on time-to-bed and LWBS by subgroup.
 - **Governance:** drift alerts/month, number of overrides, post-hoc review outcomes.
-

Data, Privacy, and Audit

- Data lineage & documentation; feature sensitivity review (pain/acuity).
 - Synthetic data **privacy tests** before any sharing; access controls and retention schedule.
-

Ownership & Next Steps

- **Now (0–4 weeks):** R1–R2 prototypes; fairness dashboard; pilot metric design.
- **Next (1–2 months):** Ward-level models (R3), tail modeling (R4), synthetic discipline (R6), queueing stub (R7).
- **Pilot (Quarter):** Shadow-mode → A/B; governance reviews; go/no-go by decision metrics.